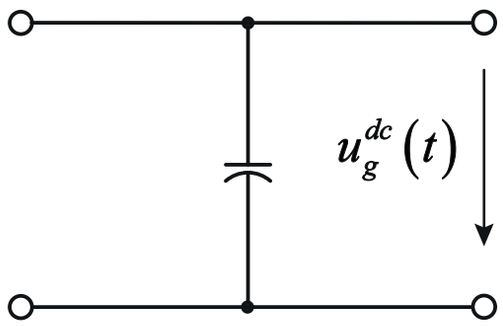
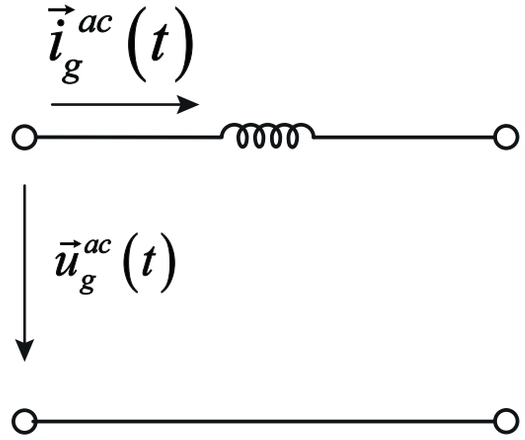
LV DC system

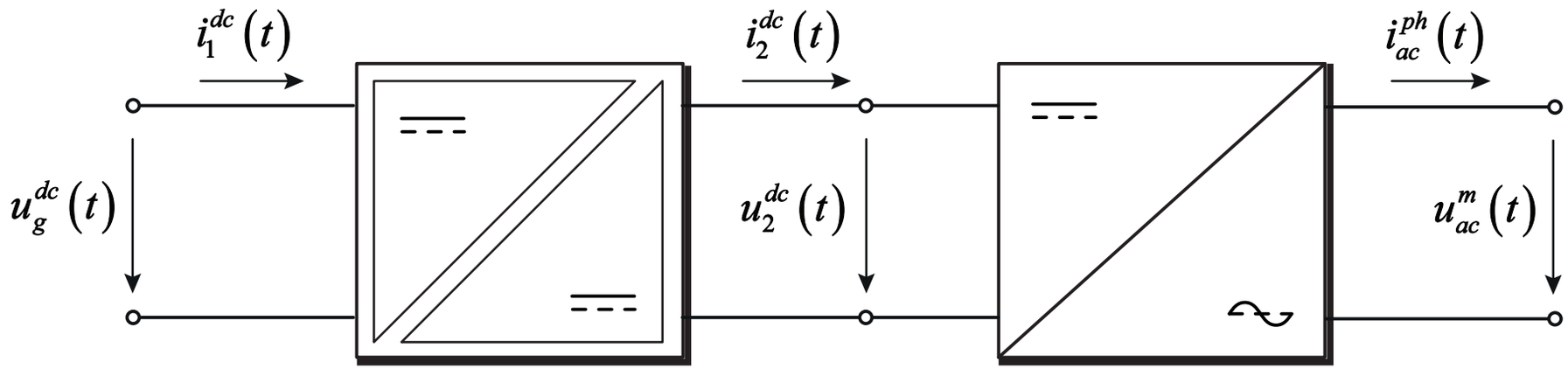
Power Flow Control Problem

MV AC 3phase system



$$\begin{cases} i_\xi^p\big|_{ref} & s.t. \quad u_g^{dc}(t) \to u_g^{dc}\big|_{ref} \\ i_\eta^p\big|_{ref} & s.t. \quad Q_g^{ac}(t) \to Q_g^{ac}\big|_{ref} \\ i_\xi^n\big|_{ref} & s.t. \quad i_\xi^n(t) \to 0 \\ i_\eta^n\big|_{ref} & s.t. \quad i_\eta^n(t) \to 0 \end{cases}$$





Fundamental isolated DC/AC module

$$i_1^{dc}\big|_{ref} \quad s.t. \quad u_2^{dc}(t) \to u_2^{dc}\big|_{ref} \qquad i_\xi^p\big|_{ref} \quad s.t. \quad u_g^{dc}(t) \to u_g^{dc}\big|_{ref}$$

$$\varphi_{dab}(t) \quad s.t. \quad i_1^{dc}(t) \to i_1^{dc}\big|_{ref}$$

$$\vec{u}_g^{ac}(t) = N\left[u_{ac}^m\big|_u(t) \quad u_{ac}^m\big|_v(t) \quad u_{ac}^m\big|_w(t)\right]^T$$

$$\vec{i}_g^{ac}(t) = N\left[i_{ac}^{ph}\big|_u(t) \quad i_{ac}^{ph}\big|_v(t) \quad i_{ac}^{ph}\big|_w(t)\right]^T$$

- *DAB works as current source*
- *Single Phase Inverter, master module works as current source*
- *Single Phase Inverter, slave modules, work as voltage source*